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# **Project Milestone #1 - Proposal**

**SFWRTECH 4SA3 – Software Architecture**

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## **Learning Objectives**

- Design and document a software architecture using the 4+1 view model.
- Implement a software architecture, including design patterns.
- Present a software architecture design.

## **Spendly – Smart Expense Tracker**

### **Description:**

Spendly is a smart yet simple expense tracking software that will be designed to help users record, organize, and review their personal spending. The purpose of the software is to allow users to log daily expenses, categorize them, keep track of spending over time, and view summarized reports such as monthly or category-based spending totals, including simple expected income and expense summaries based on recurring transactions. Spendly will focus on providing a minimal and easy-to-use system that supports basic accounting needs without unnecessary complexity, allowing non-business users to better understand their spending habits and manage their finances more effectively.

### **Target Audience:**

The target audience for Spendly is non-accounting users such as students, working professionals, and individuals who manage personal finances on a regular income. The software is intended for people who want a simple way to track their expenses, understand where their money is going, and stay aware of upcoming income and recurring costs without needing accounting knowledge or complex financial tools.

### **Functionalities and Features:**

Spendly will provide users with a set of basic functionalities focused on tracking and understanding their personal finances. This software will allow users to record their day-to-day expenses by entering details such as amount they spent, date the transaction happened, and category of that expense. Users will be able to view, update, and remove previously recorded expenses as needed.

Spendly will also support categorizing expenses to help users better understand their spending patterns. In addition, the software will generate simple summaries and reports, such as monthly or category-based spending totals. Users will also be able to define both recurring expected income and expenses, allowing the system to display expected incoming and outgoing amounts for a given time period.

## Design Pattern Opportunities:

- **MVC Design Pattern:** There are several areas within Spendly where design patterns could be applied in desirable ways to improve structure and flexibility. One potential design pattern is the Model–View–Controller (MVC) pattern, which can be used to separate expense data and business logic from user interactions and output. In this approach, the model would represent expense and income data, the view would handle displaying summaries and reports to the user, and the controller would manage user actions like adding expenses or updating that data.
- **Strategy Design Pattern:** Strategy design pattern could be used to support different ways of calculating financial summaries within the application. For example: Total monthly spendings, category-based summaries, or expected income and expense calculations based on recurring transactions. By using the Strategy pattern, these calculation methods can be kept separate and selected as needed without changing the core application logic.

*Reference: <https://www.geeksforgeeks.org/system-design/strategy-pattern-set-1/>*

## Technologies to be Used:

- **Python:** Spendly is planned to be developed using Python as the primary programming language, as it is well-suited for building small to medium-sized applications and supports clear and readable code. Python will be used to implement the core application logic, handle user input, and generate summaries and reports.
- **Firebase Firestore:** The software will use a cloud-based database, specifically Firebase Firestore, which will be responsible for persisting user data so that information remains available across sessions and can be updated over time. It allows the application to easily retrieve, add, modify, and delete financial records without requiring complex database management or SQL queries. Using Firestore also supports flexible data storage, which is suitable for a simple and evolving expense tracking application like Spendly.
- **API Use:** In addition, Spendly will integrate a third-party web service, such as a currency exchange rate API, to support basic currency conversion when displaying expense summaries. This service will allow the application to retrieve upto date exchange rates when needed.

*Reference: <https://publicapis.io/blog/currency-exchange-apis>*